

Clase 2 - Contenedores

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Archivo de Vagrant

```
Vagrant.configure("2") do |config|
  config.vm.define "server" do |server|
    server.vm.box = "generic/rocky9"
    server.vm.hostname = "server"
    server.vm.box_version = "4.3.12"

    server.vm.network "private_network", ip: "192.168.40.254", virtualbox____intnet: true

    server.vm.provider "virtualbox" do |vb|
      vb.name = "rocky9-server"
      vb.memory = 2048
      vb.cpus = 2
    end

    server.vm.provision "shell", path: "bootstrap.sh", run:
      "once"
    end
  end
end
```

Archivo de docker

```
#!/bin/bash
set -e

echo "==> Verificando si Docker ya está instalado..."

if command -v docker >/dev/null 2>&1; then
```

```
    echo "==> Docker ya está instalado. No se requiere ninguna
acción."
    exit 0
fi

echo "==> Instalando dependencias necesarias..."
dnf -y install dnf-plugins-core

echo "==> Agregando repositorio oficial de Docker..."
dnf config-manager --add-repo https://download.docker.com/linux/centos/docker-ce.repo

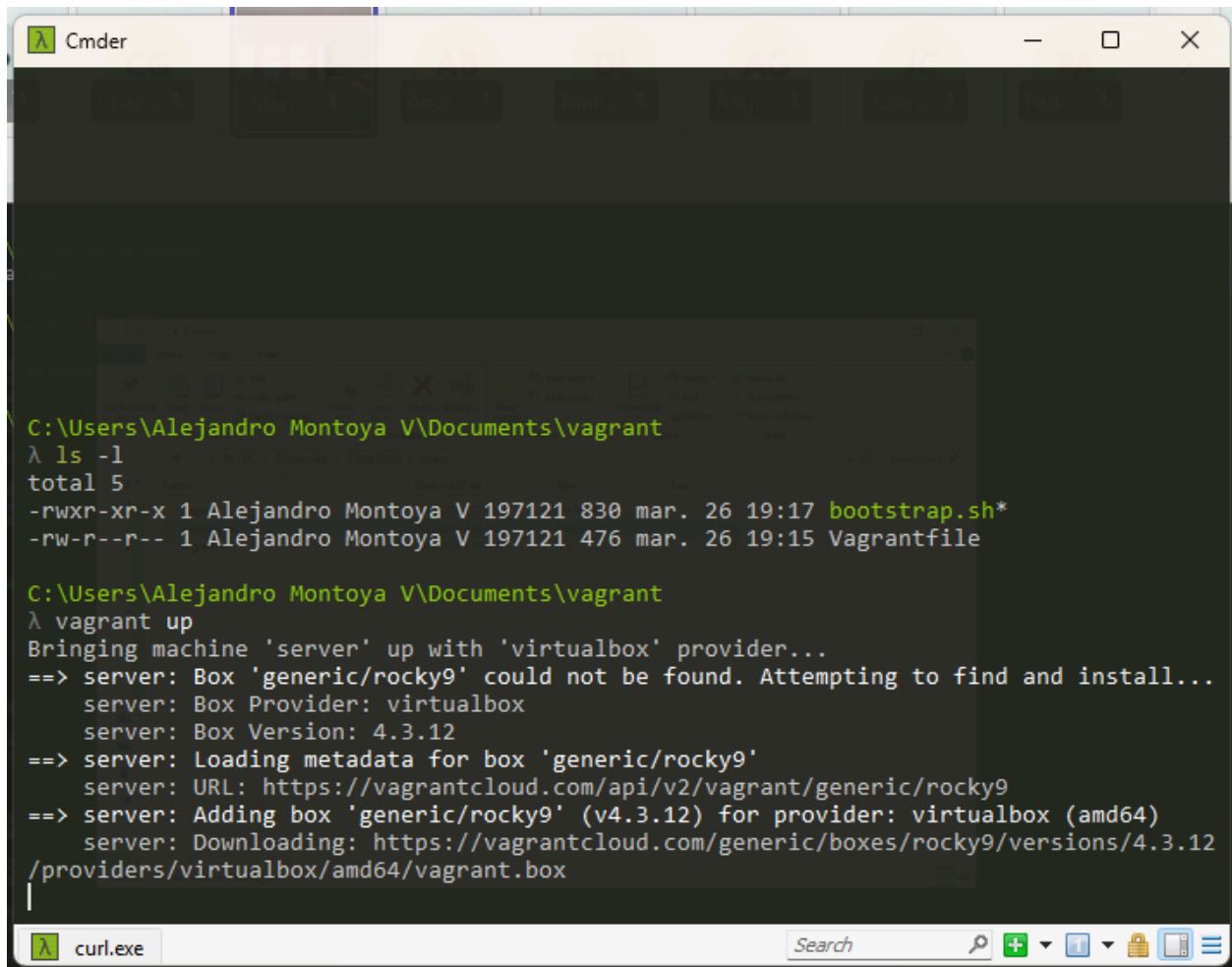
echo "==> Instalando Docker Engine..."
dnf -y install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin

echo "==> Habilitando e iniciando el servicio Docker..."
systemctl enable --now docker

echo "==> Agregando usuario vagrant al grupo docker..."
usermod -aG docker vagrant

echo "==> Verificando instalación..."
docker --version || true
docker compose version || true
```

Corremos con vagrant up

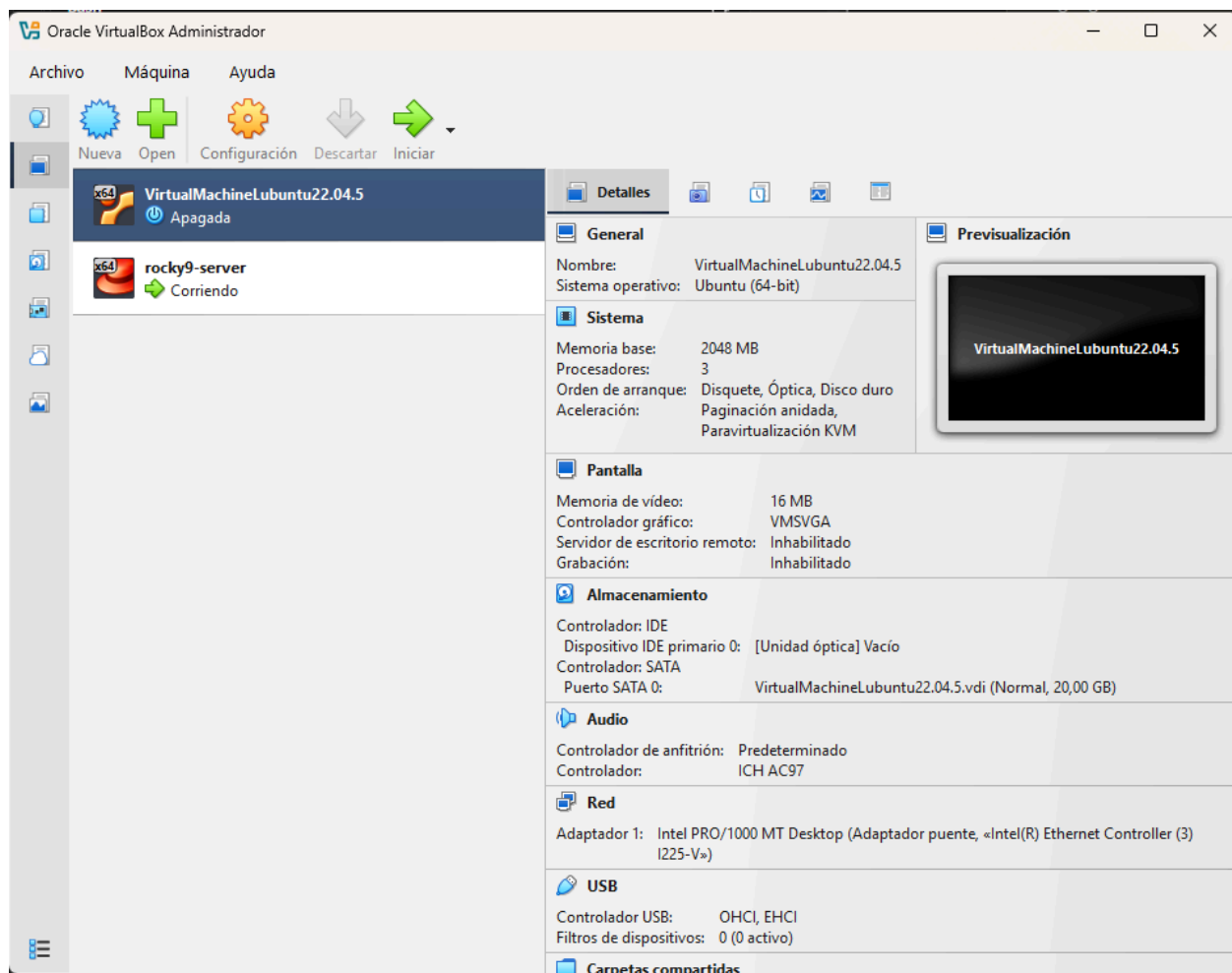


```
C:\Users\Alejandro Montoya V\Documents\vagrant
λ ls -l
total 5
-rwxr-xr-x 1 Alejandro Montoya V 197121 830 mar. 26 19:17 bootstrap.sh*
-rw-r--r-- 1 Alejandro Montoya V 197121 476 mar. 26 19:15 Vagrantfile

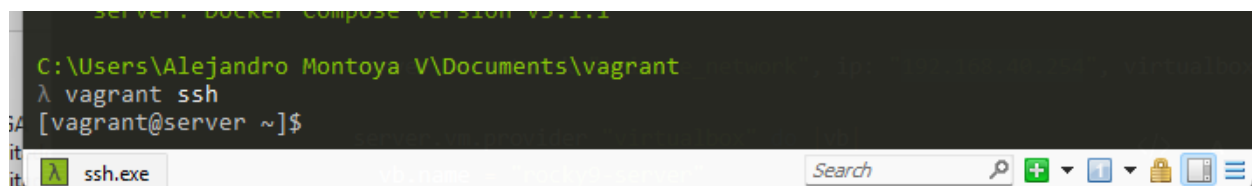
C:\Users\Alejandro Montoya V\Documents\vagrant
λ vagrant up
Bringing machine 'server' up with 'virtualbox' provider...
==> server: Box 'generic/rocky9' could not be found. Attempting to find and install...
    server: Box Provider: virtualbox
    server: Box Version: 4.3.12
==> server: Loading metadata for box 'generic/rocky9'
    server: URL: https://vagrantcloud.com/api/v2/vagrant/generic/rocky9
==> server: Adding box 'generic/rocky9' (v4.3.12) for provider: virtualbox (amd64)
    server: Downloading: https://vagrantcloud.com/generic/boxes/rocky9/versions/4.3.12
    /providers/virtualbox/amd64/vagrant.box
```

```
Cinder
server: Installing : fuse-common-3.10.2-9.el9.x86_64 11/17
server: Installing : fuse3-3.10.2-9.el9.x86_64 12/17
server: Installing : fuse-overlayfs-1.10-1.el9.7.x86_64 13/17
server: Running scriptlet: fuse-overlayfs-1.10-1.el9.7.x86_64 13/17
server: Installing : docker-ce-rootless-extras-29.3.1-1.el9.x86_64 14/17
server: Running scriptlet: docker-ce-rootless-extras-29.3.1-1.el9.x86_64 14/17
server: Installing : docker-ce-3.29.3-1.el9.x86_64 15/17
server: Running scriptlet: docker-ce-3.29.3-1.el9.x86_64 15/17
server: Running scriptlet: selinux-policy-38.1.23-1.el9.noarch 16/17
server: Cleanup : selinux-policy-targeted-38.1.23-1.el9.noarch 16/17
server: Running scriptlet: selinux-policy-targeted-38.1.23-1.el9.noarch 17/17
server: Running scriptlet: selinux-policy-targeted-38.1.05-1.el9.7.1.noarch 17/17
server: Running scriptlet: container-selinux-4:2.240.0-3.el9.7.noarch 17/17
server: Running scriptlet: passt-selinux-0*20250512.glibc1341-4.el9.7.noarch 17/17
server: Verifying : selinux-policy-targeted-38.1.23-1.el9.noarch 17/17
server: Verifying : containerd-1.0-2.2-1.el9.x86_64 1/17
server: Verifying : docker-buildx-plugin-0.31.1-1.el9.x86_64 2/17
server: Verifying : docker-ce-3.29.3-1.el9.x86_64 3/17
server: Verifying : docker-ce-cli-3.29.3-1.el9.x86_64 4/17
server: Verifying : docker-ce-rootless-extras-29.3.1-1.el9.x86_64 5/17
server: Verifying : docker-compose-plugin-5.1.1-1.el9.x86_64 6/17
server: Verifying : fuse-common-3.10.2-9.el9.x86_64 7/17
server: Verifying : container-selinux-4:2.240.0-3.el9.7.noarch 8/17
server: Verifying : fuse-overlayfs-1.10-1.el9.7.x86_64 9/17
server: Verifying : fuse3-3.10.2-9.el9.x86_64 10/17
server: Verifying : fuse3-libs-3.10.2-9.el9.x86_64 11/17
server: Verifying : passt-0*20250512.glibc1341-4.el9.7.x86_64 12/17
server: Verifying : passt-selinux-0*20250512.glibc1341-4.el9.7.noarch 13/17
server: Verifying : selinux-policy-38.1.05-1.el9.7.1.noarch 14/17
server: Verifying : selinux-policy-38.1.23-1.el9.noarch 15/17
server: Verifying : selinux-policy-targeted-38.1.05-1.el9.7.1.noarch 16/17
server: Verifying : selinux-policy-targeted-38.1.23-1.el9.noarch 17/17
server: Upgraded:
server: selinux-policy-38.1.05-1.el9.7.1.noarch
server: selinux-policy-targeted-38.1.05-1.el9.7.1.noarch
server: Installed:
server: container-selinux-4:2.240.0-3.el9.7.noarch
server: containerd-1.0-2.2-1.el9.x86_64
server: docker-buildx-plugin-0.31.1-1.el9.x86_64
server: docker-ce-3.29.3-1.el9.x86_64
server: docker-ce-cli-3.29.3-1.el9.x86_64
server: docker-ce-rootless-extras-29.3.1-1.el9.x86_64
server: docker-compose-plugin-5.1.1-1.el9.x86_64
server: fuse-common-3.10.2-9.el9.x86_64
server: fuse-overlayfs-1.10-1.el9.7.x86_64
server: fuse3-3.10.2-9.el9.x86_64
server: fuse3-libs-3.10.2-9.el9.x86_64
server: passt-0*20250512.glibc1341-4.el9.7.x86_64
server: passt-selinux-0*20250512.glibc1341-4.el9.7.noarch
server: Complete!
server: ==> Habilitando e iniciando el servicio Docker...
server: Created symlink /etc/systemd/system/multi-user.target.wants/docker.service → /usr/lib/systemd/system/docker.service.
server: ==> Agregando usuario vagrant al grupo docker...
server: ==> Verificando instalación
server: Docker version 29.3.1, build c2b9cc
server: Docker compose version V5.1.1
C:\Users\Alejandro Montoya V\Documents\vagrant
3 |
cmd.exe
```

Y vemos que la esta corriendo en virtualbox



Vemos que nos podemos conectar por ssh



Laboratorio

Por comodidad lo haremos desde wsl

0. Verificación

```
alejandro@DESKTOP-PHUCDGV:~$ docker version
Client:
 Version:           29.3.0
 API version:       1.54
 Go version:        go1.25.7
 Git commit:        5927d80
 Built:             Thu Mar  5 14:24:57 2026
 OS/Arch:           linux/amd64
 Context:           default

Server: Docker Desktop 4.66.0 (222299)
Engine:
 Version:           29.3.0
 API version:       1.54 (minimum version 1.40)
 Go version:        go1.25.7
 Git commit:        83bca51
 Built:             Thu Mar  5 14:25:43 2026
 OS/Arch:           linux/amd64
 Experimental:      false
containerd:
 Version:           v2.2.1
 GitCommit:         dea7da592f5d1d2b7755e3a161be07f43fad8f75
runc:
 Version:           1.3.4
 GitCommit:         v1.3.4-0-gd6d73eb8
docker-init:
 Version:           0.19.0
 GitCommit:         de40ad0
```

1. Docker image

1.1 Descargar una imagen

```

alejandro@DESKTOP-PHUCDGV:~$ docker pull nginx:latest
latest: Pulling from library/nginx
5e815e07e569: Pull complete
ec781dee3f47: Pull complete
bb3d0aa29654: Pull complete
cde7a05ae428: Pull complete
3189680c601f: Pull complete
587e3d84dbb5: Pull complete
510ddf6557d6: Pull complete
669e0ab8e7fa: Download complete
96a6cfe061e0: Download complete
Digest: sha256:7150b3a39203cb5bee612ff4a9d18774f8c7caf6399d6e8985e97e28eb751c18
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
alejandro@DESKTOP-PHUCDGV:~$ docker images

```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
hello-world:latest	85404b3c5395	25.9kB	9.52kB	U
mysql:8	da906917ca4a	1.09GB	248MB	U
nginx:latest	7150b3a39203	240MB	65.8MB	
postgres:15.1-alpine	f19eede5a214	347MB	94.1MB	U

```

alejandro@DESKTOP-PHUCDGV:~$

```

1.2 Construir una imagen desde Dockerfile

Crear carpeta

```

alejandro@DESKTOP-PHUCDGV:~$ mkdir docker-lab
alejandro@DESKTOP-PHUCDGV:~$ cd docker-lab/
alejandro@DESKTOP-PHUCDGV:~/docker-lab$ FROM alpine:3.20

```

Crear dockerfile

```

alejandro@DESKTOP-PHUCDGV:~/docker-lab$ nano Dockerfile
alejandro@DESKTOP-PHUCDGV:~/docker-lab$ docker build -t miapp:v1

```

Crear imagen

```

alejandros@DESKTOP-PHUCDGV:~/docker-lab$ docker build -t miapp:v1 .
[+] Building 3.1s (8/8) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile              0.0s
=> => transferring dockerfile: 147B                             0.0s
=> [internal] load metadata for docker.io/library/alpine:3.20   1.6s
=> [auth] library/alpine:pull token for registry-1.docker.io    0.0s
=> [internal] load .dockerignore                                0.0s
=> => transferring context: 2B                                    0.0s
=> [1/3] FROM docker.io/library/alpine:3.20@sha256:a4f4213abb84c497377b8544c81b3564f313746700372ec4fe84653e4fb03 0.7s
=> => resolve docker.io/library/alpine:3.20@sha256:a4f4213abb84c497377b8544c81b3564f313746700372ec4fe84653e4fb03 0.0s
=> => sha256:76eb174b37c3e263a212412822299b58d4098a7f96715f18c7eb6932c98b7efd 3.63MB / 3.63MB 0.5s
=> => extracting sha256:76eb174b37c3e263a212412822299b58d4098a7f96715f18c7eb6932c98b7efd 0.1s
=> [2/3] WORKDIR /app                                           0.0s
=> [3/3] RUN echo "Hola desde Docker" > /app/mensaje.txt      0.4s
=> exporting to image                                           0.2s
=> => exporting layers                                           0.1s
=> => exporting manifest sha256:75feab3654906eb1bc23634b26602283448d60c23042f6e8e577040e4ab05c9f 0.0s
=> => exporting config sha256:e7a017b622cee218ab04b077c2b5e2438090aa95c1efb23cf0f994bb722e48d9 0.0s
=> => exporting attestation manifest sha256:acd1def4ea70e7f2e834c3c5c3f9acbd28ec42164834ed0bd0fb3a497e2a29b2 0.0s
=> => exporting manifest list sha256:62b583c5094ebf37c0f1328deb2885570ac653b452143ec6248575ab39ec41f 0.0s
=> => naming to docker.io/library/miapp:v1                      0.0s
=> => unpacking to docker.io/library/miapp:v1                   0.0s
alejandros@DESKTOP-PHUCDGV:~/docker-lab$

```

```

alejandros@DESKTOP-PHUCDGV:~/docker-lab$ docker images

```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
hello-world:latest	85404b3c5395	25.9kB	9.52kB	U
miapp:v1	62b583c5094e	12.1MB	3.63MB	
mysql:8	da906917ca4a	1.09GB	248MB	U
nginx:latest	7150b3a39203	240MB	65.8MB	
postgres:15.1-alpine	f19eede5a214	347MB	94.1MB	U

```

alejandros@DESKTOP-PHUCDGV:~/docker-lab$ docker images | grep miapp
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
miapp:v1      62b583c5094e      12.1MB      3.63MB
alejandros@DESKTOP-PHUCDGV:~/docker-lab$

```

Ejecutamos

```

alejandros@DESKTOP-PHUCDGV:~/docker-lab$ docker run --rm miapp:v1
Hola desde Docker
alejandros@DESKTOP-PHUCDGV:~/docker-lab$

```

1.3 Etiquetar una imagen

```

alejandros@DESKTOP-PHUCDGV:~/docker-lab$ docker images | grep miapp
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
miapp:latest      62b583c5094e      12.1MB      3.63MB
miapp:v1          62b583c5094e      12.1MB      3.63MB

```

1.4 Inspeccionar una imagen

```

alejandros@DESKTOP-PHUCDGV:~$ docker image inspect miapp:v1 --format '{{.Id}}'
sha256:62b583c5094ebf37c0f1328deb2885570ac653b452143ec6248575ab39ec41f
alejandros@DESKTOP-PHUCDGV:~$

```

1.5 Guardar una imagen a tar

```

alejandro@DESKTOP-PHUCDGV:~$ docker save -o miapp.tar miapp:v1
alejandro@DESKTOP-PHUCDGV:~$ ls -lh miapp.tar
-rw----- 1 alejandro alejandro 3.5M Apr 14 08:49 miapp.tar
alejandro@DESKTOP-PHUCDGV:~$ |

```

1.6 Cargar una imagen desde tar

```

alejandro@DESKTOP-PHUCDGV:~$ docker rmi miapp:v1
Untagged: miapp:v1
alejandro@DESKTOP-PHUCDGV:~$ docker load -i miapp.tar
Loaded image: miapp:v1
alejandro@DESKTOP-PHUCDGV:~$ docker images | grep miapp
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
miapp:latest          62b583c5094e      12.1MB      3.63MB
miapp:v1              62b583c5094e      12.1MB      3.63MB
alejandro@DESKTOP-PHUCDGV:~$ |

```

1.7 Eliminar una imagen

```

alejandro@DESKTOP-PHUCDGV:~$ docker rmi miapp:latest
Untagged: miapp:latest
alejandro@DESKTOP-PHUCDGV:~$ docker images | grep miapp
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
miapp:v1              62b583c5094e      12.1MB      3.63MB
alejandro@DESKTOP-PHUCDGV:~$ |

```

1.8 Limpiar imágenes no usadas

```

alejandro@DESKTOP-PHUCDGV:~$ docker image prune -f
Total reclaimed space: 0B
alejandro@DESKTOP-PHUCDGV:~$ |

```

2. Docker container

2.1 Ejecutar un contenedor

```

alejandro@DESKTOP-PHUCDGV:~$ docker run -d --name web1 nginx:latest
efb079874e00b70fe4fd24330148c98ddd9fd6a7f9d25c1b073f2f9b0a012cd
alejandro@DESKTOP-PHUCDGV:~$ docker ps

```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
efb079874e00	nginx:latest	"/docker-entrypoint..."	3 seconds ago	Up 2 seconds	80/tcp	web1

```

alejandro@DESKTOP-PHUCDGV:~$ |

```

2.2 Ejecutar contenedor con nombre y puertos

```

alejandros@DESKTOP-PHUCDGV:~$ docker run -d --name web2 -p 8080:80 nginx:latest
f53c045408426769aaec4bb5a124bfba318c9a3a98b0885d12870e27f662bdd8
alejandros@DESKTOP-PHUCDGV:~$ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                               NAMES
f53c04540842   nginx:latest "/docker-entrypoint..." 3 seconds ago  Up 2 seconds  0.0.0.0:8080->80/tcp, [::]:8080->80/tcp  web2
efb079874e00   nginx:latest "/docker-entrypoint..." 38 seconds ago  Up 37 seconds  80/tcp                             web1
alejandros@DESKTOP-PHUCDGV:~$

```



Welcome to nginx!

If you see this page, nginx is successfully installed and working. Further configuration is required for the web server, reverse proxy, API gateway, load balancer, content cache, or other features.

For online documentation and support please refer to nginx.org.
To engage with the community please visit community.nginx.org.
For enterprise grade support, professional services, additional security features and capabilities please refer to fs.com/nginx.

Thank you for using nginx.

2.3 Listar todos los contenedores

```

alejandros@DESKTOP-PHUCDGV:~$ docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                               NAMES
f53c04540842   nginx:latest "/docker-entrypoint..." About a minute ago  Up About a minute  0.0.0.0:8080->80/tcp, [::]:8080->80/tcp  web2
efb079874e00   nginx:latest "/docker-entrypoint..." 2 minutes ago     Up 2 minutes      80/tcp                             web1
bef4d231780d   mysql:8   "docker-entrypoint.s..." 2 weeks ago      Exited (0) 6 days ago                                     mysql-container
alejandros@DESKTOP-PHUCDGV:~$

```

2.4 Ver logs

```

alejandros@DESKTOP-PHUCDGV:~$ docker logs web2
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/04/14 14:04:58 [notice] 1#1: using the "epoll" event method
2026/04/14 14:04:58 [notice] 1#1: nginx/1.29.7
2026/04/14 14:04:58 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/04/14 14:04:58 [notice] 1#1: OS: Linux 6.6.87.2-microsoft-standard-WSL2
2026/04/14 14:04:58 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/04/14 14:04:58 [notice] 1#1: start worker processes
2026/04/14 14:04:58 [notice] 1#1: start worker process 29
2026/04/14 14:04:58 [notice] 1#1: start worker process 30
2026/04/14 14:04:58 [notice] 1#1: start worker process 31
2026/04/14 14:04:58 [notice] 1#1: start worker process 32
2026/04/14 14:04:58 [notice] 1#1: start worker process 33
2026/04/14 14:04:58 [notice] 1#1: start worker process 34
2026/04/14 14:04:58 [notice] 1#1: start worker process 35
2026/04/14 14:04:58 [notice] 1#1: start worker process 36
2026/04/14 14:04:58 [notice] 1#1: start worker process 37
2026/04/14 14:04:58 [notice] 1#1: start worker process 38
2026/04/14 14:04:58 [notice] 1#1: start worker process 39
2026/04/14 14:04:58 [notice] 1#1: start worker process 40
2026/04/14 14:04:58 [notice] 1#1: start worker process 41
2026/04/14 14:04:58 [notice] 1#1: start worker process 42
2026/04/14 14:04:58 [notice] 1#1: start worker process 43
2026/04/14 14:04:58 [notice] 1#1: start worker process 44
172.17.0.1 - - [14/Apr/2026:14:05:22 +0000] "GET / HTTP/1.1" 200 896 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36" "-"
2026/04/14 14:05:22 [error] 29#29: *1 open() "/usr/share/nginx/html/favicon.ico" failed (2: No such file or directory), client: 172.17.0.1, server: localhost, request: "GET /favicon.ico HTTP/1.1", host: "localhost:8080", referer: "http://localhost:8080/"
172.17.0.1 - - [14/Apr/2026:14:05:22 +0000] "GET /favicon.ico HTTP/1.1" 404 555 "http://localhost:8080/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36" "-"
alejandros@DESKTOP-PHUCDGV:~$

```

2.5 Inspeccionar un contenedor

```

alejandros@DESKTOP-PHUCDGV:~$ docker inspect web2 --format '{{.State.Status}}'
running
alejandros@DESKTOP-PHUCDGV:~$ |

```

2.6 Detener y arrancar contenedor

```

alejandros@DESKTOP-PHUCDGV:~$ docker stop web2
web2
alejandros@DESKTOP-PHUCDGV:~$ docker start web2
web2
alejandros@DESKTOP-PHUCDGV:~$ docker ps

```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
f53c04540842	nginx:latest	"/docker-entrypoint..."	3 minutes ago	Up 3 seconds	0.0.0.0:8080->80/tcp, [::]:8080->80/tcp	web2
efb079874e00	nginx:latest	"/docker-entrypoint..."	4 minutes ago	Up 4 minutes	80/tcp	web1

2.7 Entrar en modo interactivo

```

alejandros@DESKTOP-PHUCDGV:~$ docker run -it --name alpine1 alpine:3.20 sh
Unable to find image 'alpine:3.20' locally
3.20: Pulling from library/alpine
Digest: sha256:a4f4213abb84c497377b8544c81b3564f313746700372ec4fe84653e4fb03805
Status: Downloaded newer image for alpine:3.20
/ # echo "prueba interactiva"
prueba interactiva
/ # cat /etc/os-release
NAME="Alpine Linux"
ID=alpine
VERSION_ID=3.20.9
PRETTY_NAME="Alpine Linux v3.20"
HOME_URL="https://alpinelinux.org/"
BUG_REPORT_URL="https://gitlab.alpinelinux.org/alpine/aports/-/issues"
/ # exit
alejandros@DESKTOP-PHUCDGV:~$ docker ps -a | grep alpine1
d446d2368b12 alpine:3.20 "sh" 41 seconds ago Exited (0) 8 seconds ago alpine1
alejandros@DESKTOP-PHUCDGV:~$ |

```

2.8 Pausar y reanudar

```

alejandros@DESKTOP-PHUCDGV:~$ docker pause web2
web2
alejandros@DESKTOP-PHUCDGV:~$ docker unpause web2
web2
alejandros@DESKTOP-PHUCDGV:~$ docker inspect web2 --format '{{.State.Paused}}'
false
alejandros@DESKTOP-PHUCDGV:~$ |

```

2.9 Eliminar contenedores

```

alejandro@DESKTOP-PHUCDGV:~$ docker stop alpine1
alpine1
alejandro@DESKTOP-PHUCDGV:~$ docker stop web2
web2
alejandro@DESKTOP-PHUCDGV:~$ docker rm alpine1
alpine1
alejandro@DESKTOP-PHUCDGV:~$ docker rm web2
web2
alejandro@DESKTOP-PHUCDGV:~$ |

```

```

alejandro@DESKTOP-PHUCDGV:~$ docker ps -a

```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
bef4d231780d	mysql:8	"docker-entrypoint.s..."	2 weeks ago	Exited (0) 6 days ago		mysql-container

```

alejandro@DESKTOP-PHUCDGV:~$ |

```

3. Volúmenes

3.1 Crear volumen

```

alejandro@DESKTOP-PHUCDGV:~$ docker volume create datos_nginx
datos_nginx
alejandro@DESKTOP-PHUCDGV:~$ docker volume ls

```

DRIVER	VOLUME NAME
local	6dba4908b5bc8b3a216b825efb08c62d8a0746b5aa420c0aea6976618958a8ac
local	08-running-containers_pgdata
local	datos_nginx
local	my-volume

```

alejandro@DESKTOP-PHUCDGV:~$ |

```

3.2 Inspeccionar volumen

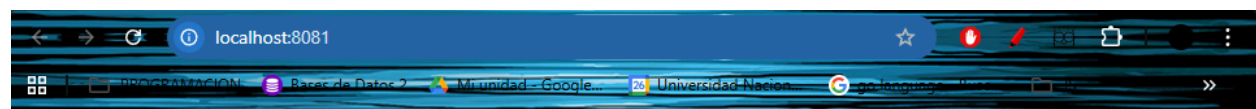
```

alejandro@DESKTOP-PHUCDGV:~$ docker volume inspect datos_nginx
[
  {
    "CreatedAt": "2026-04-14T14:23:55Z",
    "Driver": "local",
    "Labels": null,
    "Mountpoint": "/var/lib/docker/volumes/datos_nginx/_data",
    "Name": "datos_nginx",
    "Options": null,
    "Scope": "local"
  }
]
alejandro@DESKTOP-PHUCDGV:~$ |

```

3.3 Usar volumen en un contenedor

```
alejandro@DESKTOP-PHUCDGV:~$ docker run -d --name webvol -p 8081:80 -v \
datos_nginx:/usr/share/nginx/html nginx:latest
5240f72faf553821d5768c4b75bbda5c7842c72bbc742672641b6ab132775136
alejandro@DESKTOP-PHUCDGV:~$ pwd
/home/alejandro
alejandro@DESKTOP-PHUCDGV:~$ ls
docker-lab  miapp.tar  my-data
alejandro@DESKTOP-PHUCDGV:~$ echo "<h1>Hola volumen</h1>" > index.html
alejandro@DESKTOP-PHUCDGV:~$ docker cp index.html webvol:/usr/share/nginx/html/index.html
Successfully copied 2.05kB to webvol:/usr/share/nginx/html/index.html
alejandro@DESKTOP-PHUCDGV:~$ |
```



Hola volumen

3.4 Eliminar volumen

```
alejandro@DESKTOP-PHUCDGV:~$ docker rm -f webvol
webvol
alejandro@DESKTOP-PHUCDGV:~$ docker volume rm datos_nginx
datos_nginx
alejandro@DESKTOP-PHUCDGV:~$ docker volume ls
DRIVER      VOLUME NAME
local       6dba4908b5bc8b3a216b825efb08c62d8a0746b5aa420c0aea6976618958a8ac
local       08-running-containers_pgdata
local       my-volume
alejandro@DESKTOP-PHUCDGV:~$ |
```

4. Redes

4.1 Listar redes

```

alejandro@DESKTOP-PHUCDGV:~$ docker network ls
NETWORK ID          NAME                DRIVER              SCOPE
7532d2f8aa01        bridge              bridge              local
a72c34d748b0        host                host                local
3bf765d3cac0        none                null                local
alejandro@DESKTOP-PHUCDGV:~$ |

```

4.2 Crear red bridge propia

```

alejandro@DESKTOP-PHUCDGV:~$ docker network create mi_red
eaa6e37e45b5477ef519b46da618539d8251d1444d6e41ae10b0665444c10c48
alejandro@DESKTOP-PHUCDGV:~$ docker network ls
NETWORK ID          NAME                DRIVER              SCOPE
7532d2f8aa01        bridge              bridge              local
a72c34d748b0        host                host                local
eaa6e37e45b5        mi_red              bridge              local
3bf765d3cac0        none                null                local
alejandro@DESKTOP-PHUCDGV:~$ |

```

4.3 Inspeccionar red

```

alejandro@DESKTOP-PHUCDGV:~$ docker network inspect mi_red
[
  {
    "Name": "mi_red",
    "Id": "eaa6e37e45b5477ef519b46da618539d8251d1444d6e41ae10b0665444c10c48",
    "Created": "2026-04-14T16:42:06.325256704Z",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": {},
      "Config": [
        {
          "Subnet": "172.18.0.0/16",
          "Gateway": "172.18.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Options": {
      "com.docker.network.enable_ipv4": "true",
      "com.docker.network.enable_ipv6": "false"
    },
    "Labels": {},
    "Containers": {},
    "Status": {
      "IPAM": {
        "Subnets": {
          "172.18.0.0/16": {
            "IPsInUse": 3,
            "DynamicIPsAvailable": 65533
          }
        }
      }
    }
  }
]
alejandro@DESKTOP-PHUCDGV:~$

```

4.4 Conectar contenedores a una red

```

alejandro@DESKTOP-PHUCDGV:~$ docker run -d --name caja1 --network mi_red alpine:3.20 sleep 300
533c1f17c54e46a64aedd728640e574caad637b1023e4a1f14ff3e99e0e3f78e
alejandro@DESKTOP-PHUCDGV:~$ docker run -d --name caja2 --network mi_red alpine:3.20 sleep 300
cac8ea47eeb2ff66ce79fe33ac0806ba40211c029971265f46743fde3ddef69d
alejandro@DESKTOP-PHUCDGV:~$ docker exec caja1 ping -c 2 caja2
PING caja2 (172.18.0.3): 56 data bytes
64 bytes from 172.18.0.3: seq=0 ttl=64 time=0.420 ms
64 bytes from 172.18.0.3: seq=1 ttl=64 time=0.095 ms

--- caja2 ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 0.095/0.257/0.420 ms
alejandro@DESKTOP-PHUCDGV:~$ |

```

4.5 Desconectar un contenedor de la red

```

alejandro@DESKTOP-PHUCDGV:~$ docker network disconnect mi_red caja2
alejandro@DESKTOP-PHUCDGV:~$ docker exec caja1 ping -c 2 caja2
ping: bad address 'caja2'
alejandro@DESKTOP-PHUCDGV:~$ |

```

5. Docker Compose

5.1 Crear archivo compose

```

alejandro@DESKTOP-PHUCDGV:~$ nano compose.yml
alejandro@DESKTOP-PHUCDGV:~$ cat compose.yml
services:
  web:
    image: nginx:latest
    ports:
      - "8082:80"
alejandro@DESKTOP-PHUCDGV:~$ ls
compose.yml  docker-lab  index.html  miapp.tar  my-data
alejandro@DESKTOP-PHUCDGV:~$ ~|

```

5.2 Levantar servicios

```

alejandros@DESKTOP-PHUCDGV:~$ docker compose up -d
[+] up 2/2
✓Network alejandros_default Created
✓Container alejandros-web-1 Started
alejandros@DESKTOP-PHUCDGV:~$ docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED    STATUS    PORTS    NAMES
f8df7499f064   nginx:late /docker-entrypoint...    40 seconds ago    Up 39 seconds    0.0.0.0:8082->80/tcp, [::]:8082->80/tcp    alejandros-web-1
alejandros@DESKTOP-PHUCDGV:~$

```



Welcome to nginx!

If you see this page, nginx is successfully installed and working. Further configuration is required for the web server, reverse proxy, API gateway, load balancer, content cache, or other features.

For online documentation and support please refer to nginx.org.

To engage with the community please visit community.nginx.org.

For enterprise grade support, professional services, additional security features and capabilities please refer to f5.com/nginx.

Thank you for using nginx.

5.3 Ver logs

```

alejandros@DESKTOP-PHUCDGV:~$ docker compose logs
web-1 | /docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
web-1 | /docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
web-1 | /docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
web-1 | 10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
web-1 | 10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
web-1 | /docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
web-1 | /docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
web-1 | /docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
web-1 | /docker-entrypoint.sh: Configuration complete; ready for start up
web-1 | 2026/04/14 17:00:14 [notice] 1#1: using the "epoll" event method
web-1 | 2026/04/14 17:00:14 [notice] 1#1: nginx/1.29.7
web-1 | 2026/04/14 17:00:14 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
web-1 | 2026/04/14 17:00:14 [notice] 1#1: OS: Linux 6.6.87.2-microsoft-standard-WSL2
web-1 | 2026/04/14 17:00:14 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
web-1 | 2026/04/14 17:00:14 [notice] 1#1: start worker processes
web-1 | 2026/04/14 17:00:14 [notice] 1#1: start worker process 29
web-1 | 2026/04/14 17:00:14 [notice] 1#1: start worker process 30
web-1 | 2026/04/14 17:00:14 [notice] 1#1: start worker process 31
web-1 | 2026/04/14 17:00:14 [notice] 1#1: start worker process 32
web-1 | 2026/04/14 17:00:14 [notice] 1#1: start worker process 33
web-1 | 2026/04/14 17:00:14 [notice] 1#1: start worker process 34
web-1 | 2026/04/14 17:00:14 [notice] 1#1: start worker process 35
web-1 | 2026/04/14 17:00:14 [notice] 1#1: start worker process 36
web-1 | 2026/04/14 17:00:14 [notice] 1#1: start worker process 37
web-1 | 2026/04/14 17:00:14 [notice] 1#1: start worker process 38
web-1 | 2026/04/14 17:00:14 [notice] 1#1: start worker process 39
web-1 | 2026/04/14 17:00:14 [notice] 1#1: start worker process 40
web-1 | 2026/04/14 17:00:14 [notice] 1#1: start worker process 41
web-1 | 2026/04/14 17:00:14 [notice] 1#1: start worker process 42
web-1 | 2026/04/14 17:00:14 [notice] 1#1: start worker process 43
web-1 | 2026/04/14 17:00:14 [notice] 1#1: start worker process 44
web-1 | 172.19.0.1 - - [14/Apr/2026:17:00:19 +0000] "GET / HTTP/1.1" 200 896 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36" "-"
web-1 | 172.19.0.1 - - [14/Apr/2026:17:00:19 +0000] "GET /favicon.ico HTTP/1.1" 404 555 "http://localhost:8082/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36" "-"
web-1 | 2026/04/14 17:00:19 [error] 29#29: *1 open() "/usr/share/nginx/html/favicon.ico" failed (2: No such file or directory), client: 172.19.0.1, server: localhost, request: "GET /favicon.ico HTTP/1.1", host: "localhost:8082", referer: "http://localhost:8082/"
alejandros@DESKTOP-PHUCDGV:~$

```

5.4 Bajar servicios

```

alejandros@DESKTOP-PHUCDGV:~$ docker compose down
[+] down 2/2
✓Container alejandros-web-1 Removed
✓Network alejandros_default Removed
alejandros@DESKTOP-PHUCDGV:~$ docker compose ps
NAME      IMAGE      COMMAND                  SERVICE    CREATED    STATUS    PORTS
alejandros@DESKTOP-PHUCDGV:~$

```

5.5 Escalar servicio

```
alejandro@DESKTOP-PHUCDGV:~$ docker compose up -d --scale web=3
[+] up 3/4
✓Network alejandro_default Created 0.1s
✓Container alejandro-web-3 Created 0.1s
✓Container alejandro-web-1 Started 0.4s
✓Container alejandro-web-2 Started 0.7s
Error response from daemon: failed to set up container networking: driver failed programming external connectivity on endpoint alejandro-web-2 (3ee258b99af33918ea33203fe0596474e89df240804e408ca8e2ec3f89dbb77b2)
: Bind for 0.0.0.0:8082 failed: port is already allocated
alejandro@DESKTOP-PHUCDGV:~$ docker compose ps
NAME                IMAGE             COMMAND                  SERVICE    CREATED         STATUS         PORTS
alejandro-web-1     nginx:latest      "/docker-entrypoint..." web        36 seconds ago  Up 36 seconds  0.0.0.0:8082->80/tcp, [::]:8082->80/tcp
alejandro@DESKTOP-PHUCDGV:~$
```

6. Dockerfile

6.1 Ejemplo

```
alejandro@DESKTOP-PHUCDGV:~$ mkdir dockerfile-test
alejandro@DESKTOP-PHUCDGV:~$ cd dockerfile-test/
alejandro@DESKTOP-PHUCDGV:~/dockerfile-test$ echo "contenido local" > archivo.txt
alejandro@DESKTOP-PHUCDGV:~/dockerfile-test$ nano Dockerfile
alejandro@DESKTOP-PHUCDGV:~/dockerfile-test$ ls
Dockerfile  archivo.txt
alejandro@DESKTOP-PHUCDGV:~/dockerfile-test$ |
```

```
alejandro@DESKTOP-PHUCDGV:~/dockerfile-test$ docker build -t dockerfile-demo --build-arg VERSION=2.0 .
[+] Building 0.9s (10/10) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile               0.0s
=> => transferring dockerfile: 344B                               0.0s
=> [internal] load metadata for docker.io/library/alpine:3.20    0.1s
=> [internal] load .dockerignore                                  0.0s
=> => transferring context: 2B                                       0.0s
=> [1/5] FROM docker.io/library/alpine:3.20@sha256:a4f4213abb84c497377b8544c81b3564f313746700372ec4fe84653e4fb03 0.0s
=> => resolve docker.io/library/alpine:3.20@sha256:a4f4213abb84c497377b8544c81b3564f313746700372ec4fe84653e4fb03 0.0s
=> [internal] load build context                                  0.0s
=> => transferring context: 54B                                       0.0s
=> CACHED [2/5] WORKDIR /app                                       0.0s
=> [3/5] COPY archivo.txt /app/archivo.txt                       0.0s
=> [4/5] RUN echo "version=2.0" > /app/version.txt               0.3s
=> [5/5] ADD archivo.txt /app/archivo2.txt                       0.0s
=> exporting to image                                             0.3s
=> => exporting layers                                              0.1s
=> => exporting manifest sha256:08c62f26051ab3c90ddffc8495e1251b60b287398bc775fb742c17c6880ce470 0.0s
=> => exporting config sha256:da998ed1ac53dfce8dbbdf0587d483ac23709990bea59c6d7c604569eddc53ae 0.0s
=> => exporting attestation manifest sha256:7d3d3aac4a3613e3f10e7b0fb36d0cacbf6b0ed4146a2706f1a61d1855b51f32 0.0s
=> => exporting manifest list sha256:bd64e05f0c7ab30d1609b6745e32fb98e667bdc219b42c8ba1918f755a688d2c 0.0s
=> => naming to docker.io/library/dockerfile-demo:latest         0.0s
=> => unpacking to docker.io/library/dockerfile-demo:latest      0.1s
alejandro@DESKTOP-PHUCDGV:~/dockerfile-test$ |
```

```

alejandro@DESKTOP-PHUCDGV:~/dockerfile-test$ docker images

```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
alpine:3.20	a4f4213abb84	12.2MB	3.71MB	
api-golang:latest	f0297e2b88a4	24.4MB	7.64MB	
api-node:latest	b34bc4de8c1c	361MB	81.5MB	
client-react-nginx:latest	6065458eabd6	113MB	26MB	
client-react-vite:latest	62509d50a619	1.61GB	410MB	
dockerfile-demo:latest	bd64e05f0c7a	12.1MB	3.63MB	
miapp:v1	62b583c5094e	12.1MB	3.63MB	
my-ubuntu-image:latest	7a78b091c0ac	245MB	80.6MB	
mysql:8	da906917ca4a	1.09GB	248MB	U
nginx:latest	7150b3a39203	240MB	65.8MB	
postgres:15.1-alpine	f19eede5a214	347MB	94.1MB	
ubuntu:22.04	ce4a593b4e32	119MB	31.5MB	

```

alejandro@DESKTOP-PHUCDGV:~/dockerfile-test$ |

alejandro@DESKTOP-PHUCDGV:~/dockerfile-test$ docker run --rm dockerfile-demo
APP_ENV=dev
contenido local
version=2.0
alejandro@DESKTOP-PHUCDGV:~/dockerfile-test$ |

```

6.2 Inspeccionar label, env y volume

```

alejandro@DESKTOP-PHUCDGV:~/dockerfile-test$ docker image inspect dockerfile-demo
[
  {
    "Id": "sha256:bd64e05f0c7ab30d1609b6745e32fb98e667bdc219b42c8ba1918f755a688d2c",
    "RepoTags": [
      "dockerfile-demo:latest"
    ],
    "RepoDigests": [
      "dockerfile-demo@sha256:bd64e05f0c7ab30d1609b6745e32fb98e667bdc219b42c8ba1918f755a688d2c"
    ],
    "Comment": "buildkit.dockerfile.v0",
    "Created": "2026-04-14T17:38:30.23837586Z",
    "Config": {
      "Env": [
        "PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin",
        "APP_ENV=dev"
      ],
      "Cmd": [
        "sh",
        "-c",
        "echo APP_ENV=$APP_ENV && cat /app/archivo.txt && cat /app/version.txt"
      ],
      "Volumes": {
        "/data": {}
      },
      "WorkingDir": "/app",
      "Labels": {
        "autor": "prueba"
      },
      "ArgsEscaped": true
    },
    "Architecture": "amd64",
    "Os": "linux",
    "Size": 3632909,
    "RootFS": {
      "Type": "layers",
      "Layers": [
        "sha256:f5b2c0456bf1080bb3f0617295df9721aa4aa27a536bba8e301c3af8d530c9cb",
        "sha256:7dde6f1b506ff168dd66552925de0736ec8565111545c313dbc905eb858dc00",
        "sha256:7cdbae587b11c5817bc6ef92b00ed5fb91dc414c3c2d3180a070834077494ca7",
        "sha256:4be78974f192cb81d73bd3251ea5f38e2bd4f6036ad88fec48a77ca9ccd7d507",
        "sha256:e520757a4cea5c57b5884e169228a5713039fcc7f64daa8ee8a4e2f07c790664"
      ]
    },
    "Metadata": {
      "LastTagTime": "2026-04-14T17:38:30.444139379Z"
    },
    "Descriptor": {
      "mediaType": "application/vnd.oci.image.index.v1+json",
      "digest": "sha256:bd64e05f0c7ab30d1609b6745e32fb98e667bdc219b42c8ba1918f755a688d2c",
      "size": 856
    },
    "Identity": {
      "Build": [

```